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Electronic quenching rate constants of KrF(*B*,*C*) and Kr_2F^*

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The laser-assisted reaction between Kr and F_2 was used to generate KrF(*B*,*C*) molecules during a 7 ns laser pulse in a stainless steel cell. The first-order decay rates of the collisionally coupled KrF(*B*) and KrF(*C*) states in $\geqslant$50 Torr of Kr buffer gas were monitored by the KrF(*B*–*X*) fluorescence in the presence of added reagents at 300 K. Two-body quenching rate constants for KrF(*B*,*C*) by diatomic and polyatomic molecules and the rare gases were determined from the pressure dependence of the first-order decay constants of KrF(*B*,*C*). The mixed three-body quenching rate constants (with Kr) for added He, Ne, and Ar were also measured. For Kr pressure above 200 Torr, $Kr_2F(4\,^2\Gamma)$ molecules were formed via three-body quenching of KrF(*B*,*C*), and the quenching rate constants for $Kr_2F(4\,^2\Gamma)$ by the same set of reagents were determined. The general trends in the quenching constants for KrF(*B*,*C*) and $Kr_2F(4\,^2\Gamma)$ are similar to those for XeCl(*B*,*C*) and $Xe_2Cl(4\,^2\Gamma)$, respectively; however, the rate constants for KrF(*B*,*C*) and $Kr_2F(4\,^2\Gamma)$ are generally larger than for the corresponding xenon chloride molecules. Quenching mechanisms for KrF(*B*,*C*) and $Kr_2F(4\,^2\Gamma)$ are discussed.

I. INTRODUCTION

Since the KrF laser, especially the electron-beam pumped KrF laser, has the highest power and highest efficiency of the rare gas halide lasers, and since it is has been proposed as an inertial confinement fusion energy driver,[1–3] understanding the KrF* formation and quenching kinetics is important.[4–9] Although a good general understanding exists for kinetic processes in the laser medium, several detailed questions remain to be answered, and solutions to these questions require isolation and study of individual elementary reactions. Many of the original measurements of the quenching of KrF(*B*) and KrF(*C*) were done under experimental conditions such that the degree of coupling between the KrF(*B*) and KrF(*C*) states was poorly defined.[6–9] The $Kr(^3P_1)$ sensitization reaction with Cl_2 or F_2 in buffer gases is a steady-state method that can be used for observing the vibrational and electronic state relaxation and quenching of KrF(*B*) and KrF(*C*) over a wide pressure range, but obtaining rate constants is not so easy.[10–13] Eden *et al.*[9(a),9(b)] generated KrF(*B*) from pulsed UV photolysis of KrF_2; the difficulty of preparing and handling KrF_2 has limited this method for systematic study of KrF(*B*,*C*) kinetics. Electronic quenching of the collisionally coupled XeCl(*B*,*C*) molecules and $Xe_2Cl(4\,^2\Gamma)$ molecules has been studied systematically by our group using the photoassisted reaction between Xe and Cl_2 to generate XeCl(*B*,*C*).[14,15] We now report a similar study of KrF(*B*,*C*) and $Kr_2F(4\,^2\Gamma)$ using the two-photon, laser-assisted reaction between Kr and F_2.[16] This technique is based on the absorption of two UV photons to promote Kr/F_2 pairs to the $Kr^+;F_2^-$ reactive potential and the rate of KrF(*B*,*C*) formation follows the laser pulse (for lasers with nanosecond pulses). Although reaction (1) generates KrF(*B*) and KrF(*C*) molecules with high vibrational energy

$$Kr+F_2+2h\nu \rightarrow KrF(B,C)+F, \tag{1}$$

these molecules acquire a Boltzmann vibrational and electronic state distribution in Kr pressures above 50–100 Torr. The KrF(*B*–*X*) emission spectra were not explicitly modeled as a function of pressure to demonstrate Boltzmann conditions. This was done previously[14] for XeCl(*B*,*C*) and the same situation holds for KrF(*B*,*C*), since the relaxation kinetics for KrF(*B*,*C*) and XeCl(*B*,*C*) are similar.[11–13,17] The KrF(*B*) and KrF(*C*) states can be treated as a single state because of the rapid collisional mixing rate by Kr($k_{B,C}=1.2\times10^{-10}$ $cm^3\,s^{-1}$).[17] Since other gases can be added to the F_2/Kr mixture, this method provides a convenient way for measuring quenching rates.[16] In our previous report,[16(a)] the KrF(*B*,*C*) decay rates in Kr were used to determine the coupled radiative lifetime and the two-body and three-body quenching rate constants by Kr,

$$KrF(B,C) \rightarrow Kr+F+h\nu, \tag{2}$$

$$KrF(B,C)+Kr \rightarrow 2Kr+F, \tag{3}$$

$$KrF(B,C)+2Kr \rightarrow Kr_2F^*+Kr. \tag{4}$$

In addition, we reported the quenching constants of KrF(*B*,*C*) by Xe and F_2 and the KrF(*B*,*C*) equilibrium constant at 300 K from the coupled radiative decay rate.[16(a)] The coupled effective lifetime at 300 K was 25±2 ns. Based on this value and τ_b (7.0 ns) and τ_c (75 ns) (Refs. 17 and 18), the KrF(*B*,*C*) equilibrium constant at 300 K must be 4–5 with KrF(*C*) as the lower energy state. This equilibrium KrF(*C*) state concentration is larger than the value used in current models of the KrF laser. The two- and three-body quenching constants in Kr were assigned as $(4.0\pm1.0)\times10^{-12}$ $cm^3\,s^{-1}$ and $(6.0\pm2.0)\times10^{-31}$ $cm^6\,s^{-1}$, respectively. The two-body quenching constant by Xe was $(1.0\pm0.2)\times10^{-10}$ $cm^3\,s^{-1}$, and the mechanism was proven to be displacement with formation of XeF*.

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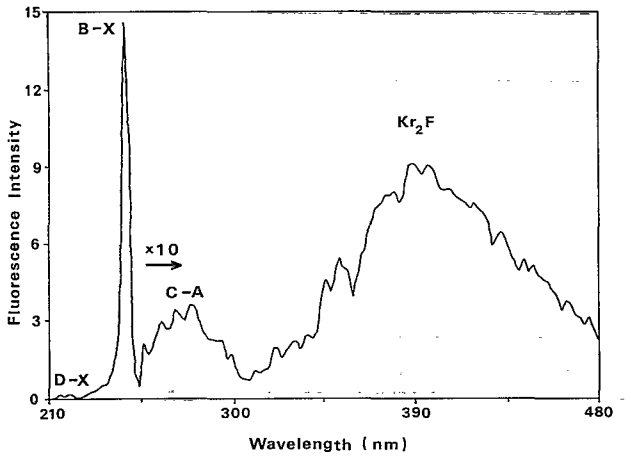


FIG. 1. The KrF(B–X), Kr(C–A), and $Kr_2F(4\,^2\Gamma$–$1,2\,^2\Gamma)$ emission spectra from excitation of Kr($2p_9$) in a mixture of 2 Torr of F_2 and 500 Torr of Kr.

In this paper we will report systematic measurements for the quenching rate constants of KrF(B,C) and $Kr_2F(4\,^2\Gamma)$ by 20 molecular reagents and rare gases. A few perfluorinated molecules, N_2, He, Ne, and Ar have small two-body rate constants, but most molecules readily quench KrF(B,C). The mixed (with Kr) three-body quenching rate constants of KrF(B,C) in He, Ne, and Ar were also measured; these can be compared to the three-body quenching rate constant in Kr.

At Kr pressure above 200 Torr, $Kr_2F(4\,^2\Gamma)$ molecules are formed via three-body quenching of KrF(B,C). Figure 1 displays the KrF(B–X), KrF(C–A), and $Kr_2F(4\,^2\Gamma$–$1,2\,^2\Gamma)$ emission spectra in 500 Torr of Kr for the convenience of the reader. The fluorescence can be used to monitor the $Kr_2F(4\,^2\Gamma)$ concentration, and a set of quenching rate constants for $Kr_2F(4\,^2\Gamma)$ were obtained. The quenching rate constants for KrF(B,C) are larger than those of $Kr_2F(4\,^2\Gamma)$; the same trend holds for XeCl(B,C) and $Xe_2Cl(4\,^2\Gamma)$.[15] In the remainder of this paper, the $4\,^2\Gamma$ state will be represented simply as Kr_2F^*. The Kr_2F^* molecules have received considerable recent attention.[19,20] Tittel and co-workers generated Xe_2Cl^* and Kr_2F^* molecules in e-beam pumping experiments with an emphasis on understanding the formation kinetics.[19] The absorption cross section for the $Kr_2F^*(9\,^2\Gamma \leftarrow 4\,^2\Gamma)$ transition has been measured;[20] there is close analogy to the Kr_2^+ absorption spectra.

This study provides a reliable set of electronic quenching rate constants for KrF(B,C) and Kr_2F^* that should be useful for modeling kinetic processes in KrF lasers. The F_2-doped crystalline Kr system is used as a model for study of charge-transfer kinetics and photodynamics in the solid state,[21] and reliable gas phase quenching constants for KrF(B,C) and Kr_2F^* also should be useful for the condensed phase. Finally, this study provides a data base for discussing the quenching mechanisms of the KrF(B,C) and Kr_2F^* molecules, which are ionic in nature and isolated from other electronic states of similar energy.

II. EXPERIMENTAL TECHNIQUES

The experiments were done in a laser-induced fluorescence cell made of stainless steel with baffle arms terminating in quartz windows mounted at the Brewster angle. The cell was constructed from a piece of 6 cm i.d. and 10 cm long pipe to which a 1.5 cm i.d. pipe was welded as baffle arms. The arms were only 10 cm long in order to minimize the volume in the baffle arms. Two quartz windows were sealed to the sides of the LIF cell for observation of the fluorescence. The laser wavelength 243 nm chosen to drive reaction (1) lies to the short wavelength side of the maximum (246 nm) in the two-photon excitation spectrum of Kr/F_2.[16(b)] This was the best compromise to permit observation of the KrF(B–X) fluorescence at 248 nm with minimum scattered laser light and still have adequate excitation efficiency. A Nd-YAG laser (Quanta Ray DCR) operated at 532 nm was used to pump a dye laser (Quanta Ray PDL-1). The dye was DCM. The dye laser output was frequency doubled and then mixed with the 1064 nm fundamental from the YAG laser in a KD*P crystal to obtain the desired UV wavelength. The laser pulse was $\approx$1 mJ with 7 ns full width at half-maximum (FWHM). This short laser pulse was needed because of the short decay time of KrF(B,C) molecules. Decay times of $\geqslant$5 ns could be reliably measured by monitoring the fluorescence signal over two decay lifetimes. The UV laser beam which was focused at the center of the cell with a 0.5 m lens passed vertically through the cell so that laser beam was aligned with the slit of the monochromator.

The KrF(B,C) fluorescence was detected using a R-2693 (Hammamatsu) photomultipier tube placed in a specially wired base; the rise time of the signal with this base was 1.5 ns. A glass filter (catalog no. 51670, Oriel Co.) plus an interference filter with peak transmission at 410 nm together with a photomultiplier tube (1P28, RCA) was placed on the other side of the cell to observe the Kr_2F^* fluorescence (see Fig. 1). The KrF(B–X) fluorescence was collected with a short focal length lens and focused onto the entrance slit of the 1 m monochromator set at 248 nm. The fluorescence signals from the photomultiplier tubes were recorded and accumulated with either a LeCroy Digitizer [1.3 G samples] or with a Hewlett Packard digital storage scope [1.0 G samples]. The waveforms were transferred to a computer and subsequently analyzed to obtain decay constants. The Kr_2F^* decay was much longer than that for KrF(B–X) and the scattered laser light and KrF(C–A) fluorescence, which was present in the early part of Kr_2F^* waveform, could be ignored in the analysis of the Kr_2F^* decay. Usually the waveforms for KrF(B–X) and Kr_2F^* fluorescence were obtained from the average of 1000 laser shots with a good signal-to-noise ratio. Figures 2(a) and 2(b) show some typical waveforms from the KrF(B–X) and Kr_2F^* fluorescence with added reagent gas.

Kr and Xe, research grade, were purchased from Cryogenic Rare Gases. Kr was quoted with a purity of 99.995% with 25 ppm Xe as the major contaminant. Tank grade He, Ne, and Ar (99.995%) were purified by passage through molecular sieve filled traps cooled to 196 K. To confirm the

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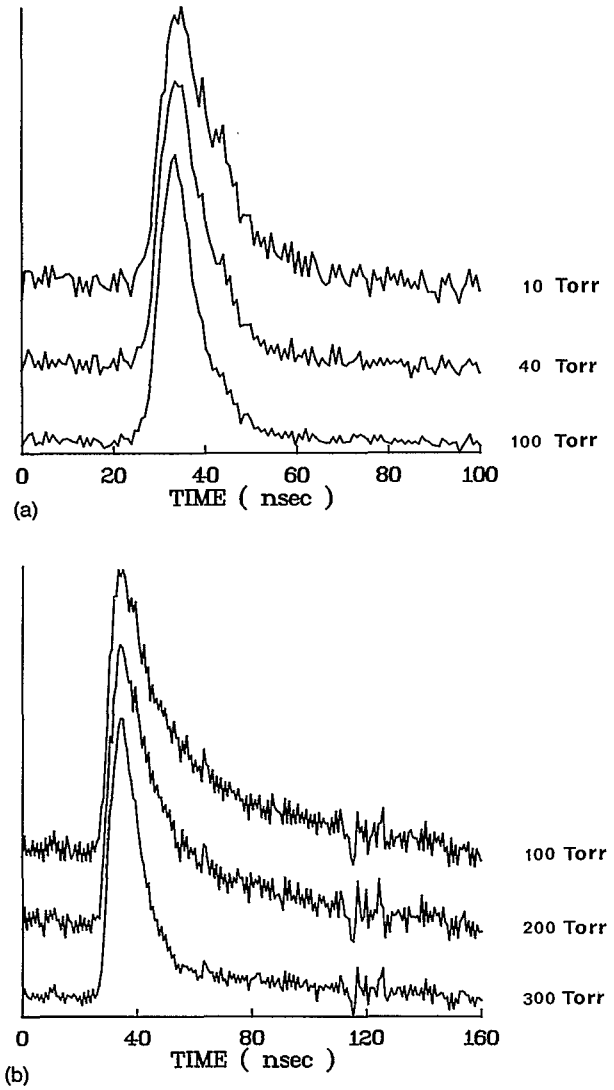


FIG. 2. Sample waveforms for the KrF(B–X) and Kr_2F^* fluorescence. (a) Decay curves for KrF(B–X) in SF_6 with 100 Torr of Kr and 1 Torr of F_2; the SF_6 pressures are indicated. (b) Decay curves for Kr_2F^* in C_2F_6 with 300 Torr of Kr and 2 Torr of F_2; the C_2F_6 pressure is indicated. The signal in the first 10 ns is KrF(B,C) emission plus scattered laser light; our analysis for the Kr_2F^* decay began after 15–20 ns.

Ar rate constant measurements, a higher purity lecture bottle of Ar (99.998%, Matheson) was purchased and used for some experiments; the same decay constants were obtained from both sources of Ar. A high purity N_2 cylinder (99.99%) was also purchased from Matheson for measuring the rate constant to avoid effects of impurities. HF was purchased for Matheson as a 7% mixture in He. Other reagent gases were degassed and purified by fractional distillation prior to use. The purified gases were loaded in Pyrex glass reservoirs, which were connected to the stainless cell. The F_2 source for most experiments was a 10% mixture in He (Cryogenic Gases). In order to confirm the F_2 quenching rate constant, data were obtained with a second tank having 5% F_2 in He (Spectra Gases, Inc.). The gas cell and gas handing lines were passivated periodically with F_2; the KrF(B–X) and Kr_2F^* fluorescence intensities for fixed gas composition were constant over a 30 min period in a passivated cell. The pressures were measured with MKS Baratron pressure transducers with 0–10 and 0–1000 Torr heads, which were calibrated with a Hg manometer.

Preparing high pressure samples for measurement of the KrF(B,C) and Kr_2F^* quenching constants must be done carefully to ensure homogenous mixtures. The reagent pressure varied from 0–10 Torr for reagents with large constants and those samples were prepared by placing the quenching gas in the cell first and then adding F_2/He and finally Kr. For the reagents with small quenching constants, such as rare gases, the required reagent pressures extend up to 400 Torr. In these cases, F_2/He was placed in the cell first, Kr was then added, and finally the reagent gas was added. Samples were prepared and allowed to mix for a half-hour before data were collected. We saw no evidence for the need of longer equilibration times, i.e., the decay constants did not change for longer mixing times. Since the Kr pressures for the KrF(B,C) and Kr_2F^* quenching measurement are quite different, the quenching rate constants for KrF(B,C) and Kr_2F^* were obtained from separate experiments even for the same reagent.

III. EXPERIMENTAL RESULTS

A. Two-body quenching constants for KrF(B,C) by molecular reagents

The KrF(B,C) molecules were generated by the two-photon (243 nm) laser-assisted reaction between Kr and F_2 in 100 Torr of Kr, except for a few experiments in 50 Torr of Kr. The F_2 pressures were 1 or 2 Torr in order to minimize the quenching of KrF(B,C) by F_2. For these conditions, the KrF(B) and KrF(C) states can be treated as a single mixed state[16,17] and the quenching constants of KrF(B,C) were determined by monitoring the decay rate of the KrF(B–X) emission in various pressures of reagent. In all cases, the waveforms for the KrF(B–X) emission could be fitted to a single exponential [see Fig. 2(a) for an example]. The three-body quenching rate of KrF(B,C) is not serious for 50–100 Torr of Kr, but quenching by F_2 and two-body quenching by Kr must be considered. The total first-order decay constant for KrF(B,C) with added reagent Q is given by

$$k=\tau_{\text{eff}}^{-1}+k_{\text{Kr}}[\text{Kr}]+k_{\text{F}_2}[\text{F}_2]+k_Q[Q], \tag{5}$$

where τ_{eff} is the effective radiative decay time of KrF(B,C). If the Kr and F_2 pressures are constant, Eq. (5) reduces to Eq. (6)

$$k=k'+k_Q[Q], \tag{6}$$

and the quenching rate constants, k_Q, can be obtained from the slope of the plot of the decay constants for KrF(B–X) emission vs the reagent pressure. Fifteen molecules were chosen for two-body rate constant measurements, and Stern–Volmer plots that are representative of reagents with fast and slow quenching rates are presented in Figs. 3 and 4. The collisionally coupled radiative lifetime is ~25 ns and the reagent pressure could be increased until the KrF(B,C) decay times were reduced to ~6 ns; therefore,

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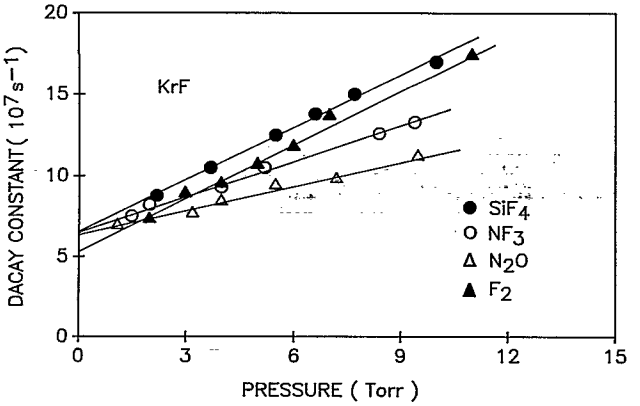


FIG. 3. Concentration dependence of the decay constants for KrF(B,C) in SiF_4, NF_3, and N_2O. The Kr and F_2 pressures were 100 and 1 Torr, respectively. The F_2 quenching data also are shown for the 5% mixture of F_2 in He as the F_2 source. All data were acquired by observing the KrF(B–X) emission at 248 nm with 1 mm slit width of the monochromator for 1000 laser shots.

reliable Stern–Volmer plots could be obtained. The rate constants from the Stern–Volmer plots generally had least-square uncertainties of ±10%. The values for k' in 50 and 100 Torr of Kr with 1 Torr of F_2 should be 5.0×10^7 and 6.2×10^7 s^{-1}, respectively, according to our previous work.[16] For 1 Torr of F_2 and 100 Torr of Kr, the intercepts were always between 6.0 and 6.5×10^7 s^{-1} and the intercepts from the current series of experiments were consistent with the k' values mentioned above. This extensive series of measurements confirms the value of τ_{eff}^{-1} for KrF(B,C), which had been recommended as ~25±2 ns.[16(a)]

Earlier measurements[6,9] of the KrF(B,C) quenching constant by F_2 gave $(5.7\pm0.5)\times10^{-10}$ and $(4.8\pm0.3)\times10^{-10}$ $cm^3\,s^{-1}$. We previously reported F_2 quenching data obtained in both 50 and 100 Torr of Kr for an extensive range of F_2 pressure using the 10% F_2 mixture as the fluorine source. The rate constant was $(2.8\pm0.2)\times10^{-10}$ $cm^3\,s^{-1}$, which agreed with less extensive measurements in our laboratory[22] using a pure F_2 source to prepare the samples. Since the correct F_2 rate constant is important, a new set of experiments was done using the 5% mixture as the F_2 source. The data from the new F_2 source are included in Fig. 3; the rate constant determined from the slope is $3.5\pm0.4\times10^{-10}$ $cm^3\,s^{-1}$. The average of all three determinations $(3.0\pm0.3)\times10^{-10}$ $cm^3\,s^{-1}$ should be reliable and it is listed in Table I. We strongly support this smaller k_{F_2} value rather than a value near 5×10^{-10} $cm^3\,s^{-1}$. Hakuta *et al.*[17] recently reported $k_{F_2} = 2.5 \pm 1.0\times10^{-10}$ $cm^3\,s^{-1}$, which agrees with our claim.

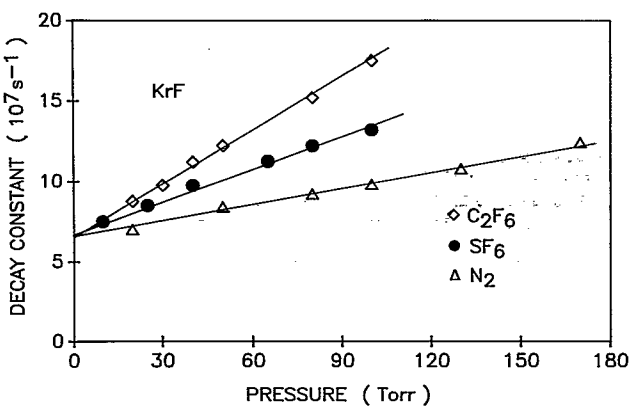


FIG. 4. Concentration dependence of the decay constants for KrF(B,C) in C_2F_6, SF_6, and N_2. The Kr and F_2 pressures were 100 and 1 Torr, respectively. The KrF(B–X) emission was observed at 248 nm with 1 mm slit width of the monochromator for 1000 laser shots.

TABLE I. Two-body quenching rate constants[a] for KrF(B,C) and $Kr_2F(4\,^2\Gamma)$.

Reagents	KrF(B,C)	$Kr_2F(4\,^2\Gamma)$
N_2	$(1.1\pm0.2)\times10^{-11}$	$(\leqslant1.0)\times10^{-13}$
CO	$(1.6\pm0.2)\times10^{-10}$	$(5.7\pm0.6)\times10^{-11}$
O_2	$(1.8\pm0.2)\times10^{-10}$	$(4.1\pm0.5)\times10^{-11}$
F_2	$(3.0\pm0.3)\times10^{-10}$ [b]	$(1.2\pm0.2)\times10^{-10}$ [b]
HF	$(3.1\pm0.4)\times10^{-10}$	$(3.7\pm0.4)\times10^{-11}$
CO_2	$(2.4\pm0.2)\times10^{-10}$	$(2.7\pm0.3)\times10^{-11}$
	7.7×10^{-10} [e]	
N_2O	$(1.2\pm0.1)\times10^{-10}$	$(3.0\pm0.3)\times10^{-11}$
NF_3	$(2.2\pm0.2)\times10^{-10}$	$(8.4\pm1.0)\times10^{-12}$
	$(5.2\pm0.5)\times10^{-11}$ [f]	5.1×10^{-11} [g]
BF_3	$(1.7\pm0.2)\times10^{-10}$	$(2.6\pm0.4)\times10^{-11}$
SiF_4	$(3.2\pm0.4)\times10^{-10}$	$(3.5\pm0.4)\times10^{-11}$
SF_6	$(2.0\pm0.3)\times10^{-11}$	$(4.8\pm1.0)\times10^{-13}$
CF_4	$(2.6\pm0.3)\times10^{-11}$	$(\leqslant1.0)\times10^{-13}$
C_2F_6	$(3.3\pm0.4)\times10^{-11}$	$(1.8\pm0.3)\times10^{-12}$
CF_3Cl	$(1.1\pm0.1)\times10^{-10}$	$(3.8\pm0.4)\times10^{-11}$
C_2F_4	$(2.5\pm0.3)\times10^{-10}$	$(3.1\pm0.3)\times10^{-11}$
He	$(2.5\pm0.5)\times10^{-12}$	$<1.0\times10^{-13}$
	$(3.3\pm1.1)\times10^{-13}$ [f]	
Ne	$(2.1\pm0.5)\times10^{-12}$	$<1.0\times10^{-13}$
	$(1.6\pm0.4)\times10^{-12}$ [f]	
Ar	$(2.9\pm0.5)\times10^{-12}$	$<1.0\times10^{-13}$
	$(1.8\pm0.6)\times10^{-12}$ [f]	
Kr	$(4.0\pm0.5)\times10^{-12}$ [c]	$<0.6\times10^{-13}$
Xe	$(1.0\pm0.2)\times10^{-10}$ [c]	$(1.1\pm0.3)\times10^{-10}$
		$(1.5\pm0.3)\times10^{-10}$ [d]

[a]In units of cm^3 $molecule^{-1}$ s^{-1}.
[b]This is the average value from our work (see the text for discussion of other measurements for both KrF* and Kr_2F^*).
[c]Taken from Ref. 17.
[d]Taken from Ref. 16.
[e]Taken from Ref. 23.
[f]Taken from Ref. 9(b).
[g]Taken from Ref. 19(b).

The two-body quenching rate constants for KrF(B,C) are listed in Table I. Except for N_2, SF_6, CF_4, and C_2F_6, all other k_Q values are $1\text{–}4\times10^{-10}$ $cm^{-3}\,s^{-1}$. We initially were surprised that so many of the k_Q values were in the $(2\text{–}4)\times10^{-10}$ $cm^3\,s^{-1}$ range. To verify these large rate constants, several checks were made by repeating the measurements using different sources of the gases. Also, the quenching data for several reagents were rechecked several months later. The results in Table I were found to be reproducible. In particular, two independent sets of data were collected for NF_3 and each gave the rate constant in Table I. Although experiments were not done with H_2O, it

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seems certain that the rate constant would be large (probably similar to that for HF). Rate constants for CO_2 and NF_3 have been measured by other workers using the photolysis of KrF_2 to generate $KrF(B)$, and these results also are included in Table I.[9,23] These rate constants were measured by monitoring the $KrF(B–X)$ fluorescence from KrF_2 photolysis without added buffer gas, and the measurements are *not* for a 300 K equilibrium mixture of B and C states. The CO_2 measurement used 159 nm photolysis, which gives vibrationally excited $KrF(B)$. The implication from the CO_2 data is that vibrationally excited $KrF(B)$ is more readily quenched than is the 300 K equilibrium mixture of $KrF(B,C)$. The NF_3 quenching measurement was done by photolysis of KrF_2 at 193 nm, which generates vibrationally cold $XeF(B)$ in the absence of $KrF(C)$. Since the radiative lifetime of $KrF(B)$ is $\sim$7 ns, observing the reduction in decay time with added reagent was not easy. Inspection of Eden's data[9] suggests that the uncertainty in k_{NF_3} is probably more than the stated $\pm 10\%$ and, if the highest $[NF_3]$ point is ignored, the rate constant would be 30% larger. Nevertheless, there is a difference between Eden *et al.* and our measurements; this difference could imply that the rate constants for quenching of $KrF(B)$ and $KrF(C)$ by NF_3 are not the same. The quenching constants for N_2, SF_6, CF_4, and C_2F_6 are in the 10^{-11} cm^3 s^{-1} range, which is an order of magnitude larger than those for rare gases (except for Xe). Replacing one F by a Cl atom in CF_4 enhances the rate constant for quenching of $KrF(B,C)$ by an order of magnitude. A similar enhancement in k_Q was found for quenching of $XeCl(B,C)$ by CF_3H and CF_3Cl relative to CF_4. The introduction of the double bond in C_2F_4 increased the quenching constant by an order of magnitude relative to C_2F_6. Several perfluorinated molecules (SiF_4, NF_3, and BF_3) have large quenching rate constants. The general trend for the quenching of $KrF(B,C)$ by stable molecules are similar to those for $XeCl(B,C)$; however, the quenching rate constants for $KrF(B,C)$ by SF_6, CF_4, and NF_3 are almost ten times larger than those for $XeCl(B,C)$.

The measurements for the two-body quenching constants by the rare gases is described below. However, we wish to emphasize here that direct, two-body quenching of $KrF(B,C)$ by rare gases does exist, and that the rate constants are $\sim$two times larger than for $XeCl(B,C)$. Of course, the Xe quenching rate constant for $KrF(B,C)$ is much larger than for $XeCl(B,C)$, because formation of $XeF(B,C)$ by displacement occurs on virtually each collision.[16(a)]

B. Two- and three-body quenching constants for KrF(B,C) by rare gases

The two-body and mixed three-body quenching rate constants for $KrF(B,C)$ in Kr with added He, Ne, and Ar were measured. The three-body process corresponds to Eq. (7)

$$KrF(B,C)+Kr+Q\rightarrow Kr_2F^*+Q. \tag{7}$$

The yields of Kr_2F^* were not measured, but the Kr_2F^* fluorescence was observed with added He, Ne, and Ar, as well as with Kr, and we have no reason to suspect a dark three-body quenching channel. The mixed three-body constant for Xe could not be measured since Xe has a very large two-body constant. The two-body constants for N_2, CF_4, and SF_6 also were too large to permit study of their three-body rates (with Kr), although these were observed with $XeCl(B,C)$.[15] Since impurities could affect the measurements at the required high pressures, the He, Ne, and Ar gases were purified via passage through a molecular sieve trap cooled to 196 K. Also a high purity tank of Ar was used to confirm the data.

The first-order decay constants for variable He, Ne, and Ar pressure were measured for Kr pressures of 50, 100, 150, and 200 Torr in order to separate the two- and three-body components. The total first-order decay constant for $KrF(B,C)$ can be expressed as

$$k=\tau_{\text{eff}}^{-1}+k_{F_2}[F_2]+k_{Kr}[Kr]+k'_{Kr}[Kr]^2 +k_Q[Q]+k_Q^{Kr}[Kr][Q]. \tag{8}$$

In Eq. (8), k_Q and k_Q^{Kr} represent the two-body and mixed three-body quenching rate constants for Q=He, Ne, and Ar. The three-body rate constants for the 2He and 2Ne processes are less than 10^{-32} cm^6 s^{-1} and can be ignored.[9(d)] The three-body constant for the 2Ar process ($2–8\times10^{-32}$ cm^3 s^{-1}) is larger (because of ArKrF* formation);[7,9,13] however, we found no evidence for the pure three-body rate in competition with the mixed Ar+Kr rate for pressures $\leqslant$400 Torr. For constant Kr and F_2 pressure, Eq. (8) reduces to Eq. (9)

$$k=k''+k_Q[Q]+k_Q^{Kr}[Kr][Q]. \tag{9}$$

The slopes of the plots of the first-order decay constants for $KrF(B)$ vs $[Q]$ give the sum of the two- and three-body rate constants $k_{\text{eff}}=k_Q+k_Q^{Kr}[Kr]$. The k_Q and k_Q^{Kr} rate constants can be separated by plotting k_{eff} vs Kr pressure and finding the slope and intercept.

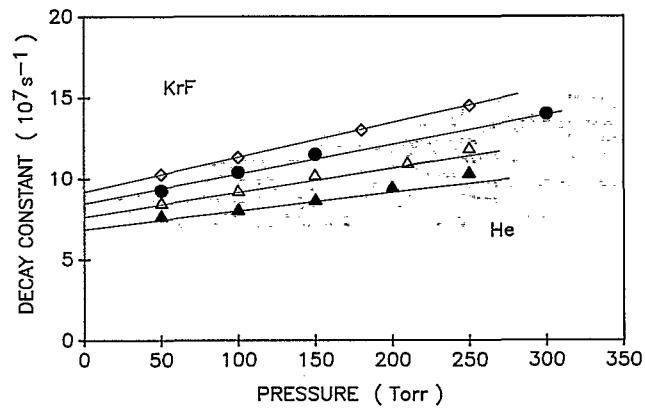


FIG. 5. Pressure dependence of the $KrF(B,C)$ decay constants in He with 2 Torr of F_2 and variable Kr: (▲) 50; (△) 100; (●) 150; and (◇) 200 Torr.

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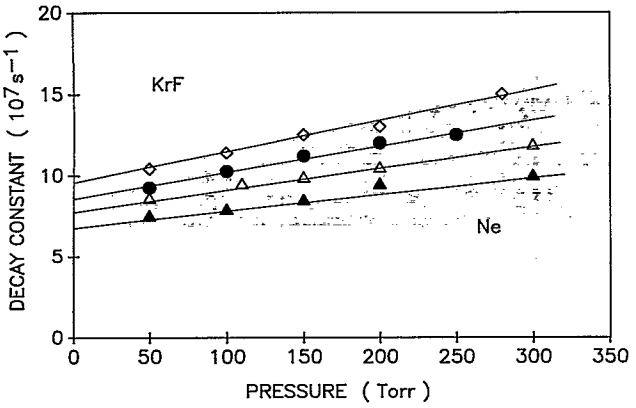


FIG. 6. Pressure dependence of the decay KrF(B,C) decay constant in Ne with 2 Torr of F_2 and variable Kr: (▲) 50; (△) 100; (●) 150; and (◇) 200 Torr.

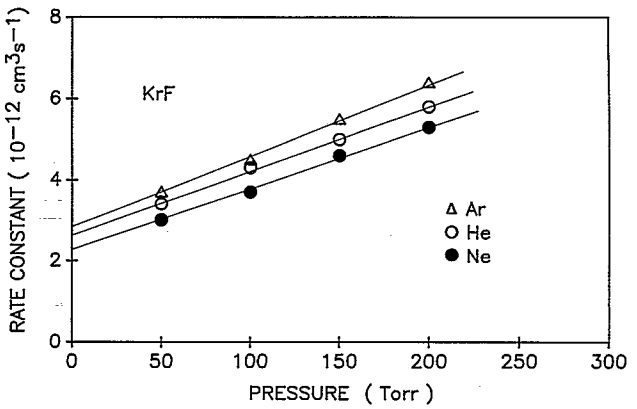


FIG. 8. Plots of effective quenching rate constants for KrF(B,C) in He, Ne, and Ar vs Kr pressure. These k_{eff} values were obtained from the slopes of the plots in Figs. 4, 5, and 6. The two-body and three-body quenching constants of Eq. (9) are given by the intercept and slope of each line.

Plots of the first-order decay constants of KrF(B,C) with added He, Ne, and Ar for 50, 100, 150, and 200 Torr of [Kr] are given in Figs. 5–7. The slopes and intercepts give k_{eff} and k'', respectively. The k'' values increase with Kr pressure in accord with the previously reported two- and three-body rate constants for Kr. The k_{eff} values from the slopes of these plots are shown vs Kr pressure in Fig. 8. The plots are linear and the two-body and mixed three-body quenching rate constants in He, Ne, and Ar were obtained from the intercept and slope of each plot. The two-body constants are listed in Table I and the three-body constants are listed in Table II. The two-body constants are $(2.5\pm0.5)\times10^{-12}$, $(2.1\pm0.5)\times10^{-12}$, and $(2.9\pm0.5)\times10^{-12}$ $cm^3 s^{-1}$, respectively, which are the same to within the experimental uncertainty. Given the very large quenching rate constants by common tank gas impurities, these two-body rate constants probably should be viewed as upper limit values. The uncertainty in these two- and three-body rate constants is about ±20%, since these values were obtained from two plots and the formal uncertainties of the slopes from each plot were ±10%. The actual uncertainty is probably even larger because of the difficulty in resolving the two- and three-body components. The mixed three-body constants are $(4.6\pm1.5)\times10^{-31}$, $(4.9\pm1.5)\times10^{-31}$, and $(5.5\pm2.0)\times10^{-31}$ $cm^6 s^{-1}$ for He, Ne, and Ar, respectively. These values are only slightly smaller than the recommended two- and three-body constants for Kr $(4.0\pm0.5)\times10^{-12}$ $cm^3 s^{-1}$ and $(6.0\pm2.0)\times10^{-31}$ $cm^6 s^{-1}$, respectively.[16(a)]

Our two-body constants are compared to those reported by Eden *et al.*[9] from photolysis of KrF_2 in Table I. The values for Ne and Ar agree to within a factor of 2, which is somewhat surprising considering that the photolysis experiments initially give only KrF(B). One implication is that two-body electronic quenching and mixing have comparable rates. The sensitization studies[10,11] in He, Ne, and Ar also suggested rather small $B\leftrightarrow C$ mixing rate constants for low v levels, and the k_{BC} values for He, Ne, and Ar may be much smaller than for Kr.[17,24] The rate constant for He derived from experiments[9] using the photolysis of KrF_2 is eight times smaller than our value; however, their plot only has four points and the degree of quenching is small. Nevertheless, our k_{He} value does seem

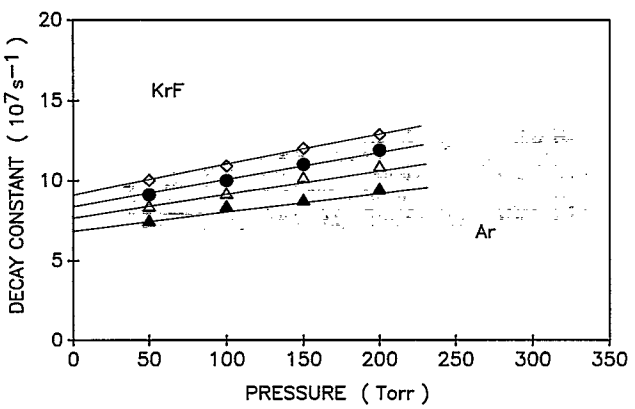


FIG. 7. Pressure dependence of the KrF(B,C) decay constants in Ar with 2 Torr of F_2 and variable Kr (▲) 50; (△) 100; (●) 150; and (◇) 200 Torr. The KrF(B–X) fluorescence for Figs. 4, 5, and 6 was observed in the same way as for Figs. 2 and 3.

TABLE II. Three-body quenching rate constants[a] for KrF(B,C) and with comparison to XeCl(B,C).

Reagents	KrF(B,C)	XeCl(B,C)[c]
He+Kr(Xe)[b]	$(4.6\pm1.5)\times10^{-31}$	$(2.4\pm0.5)\times10^{-31}$
Ne+Kr(Xe)[b]	$(4.9\pm1.5)\times10^{-31}$	$(7.4\pm1.5)\times10^{-31}$
Ar+Kr(Xe)[b]	$(5.5\pm2.0)\times10^{-31}$	$(8.9\pm1.9)\times10^{-31}$
	$\sim1.7\times10^{-31}$ [d]	
2Kr(Kr+Xe)[b]	$(6.0\pm2.0)\times10^{-31}$	$(9.9\pm1.9)\times10^{-31}$
(2Xe)[b]		$(13.0\pm4.0)\times10^{-31}$

[a]In units of cm^6 molecule^{-2} s^{-1}.
[b]The parentheses denote the rare gas for XeCl(B,C) experiments.
[c]Taken from Ref. 14.
[d]Taken from Ref. 7 and adjusted to τ_{eff}=25 ns.

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high relative to the trend normally expected from rare gases.

The mixed three-body rate constants are in order of $k_{He}^{Kr} \approx k_{Ne}^{Kr} < k_{Ar}^{Kr} < k_{Kr}^{Kr}$, even though the actual difference between them is small. Our k_{Ar}^{Kr} value is in approximate agreement with an early report by Rokni and co-workers[7] after adjustment for the currently accepted τ_{eff}. The same types of experiments have been done for XeCl(B,C), and those rate constants[14] are also listed in Table II. The largest difference between the two sets is for the parent rate constants; k_{Xe}^{Xe} is two times larger than k_{Kr}^{Kr}. The small decline in the mixed three-body rate constants for XeCl(B,C) in the Kr, Ar, and Ne species resembles that for KrF(B,C) with Ar and Ne. The three-body constant for He+Kr+KrF(B,C) may be too large, or that for He +Xe+XeCl(B,C) may be too small. However, neither set of values declines very much as the rare gas becomes lighter, and stabilization of the initially formed Kr_2F^{**} or Xe_2Cl^{**} by collisions with the chaperone is not highly dependent on the nature of the rare gas atom. The three-body quenching rates of XeCl(B,C) were nearly equal to the Xe_2Cl^* formation rate.[14,15] We did not make any quantitative measurements, but the Kr_2F^* emission intensities with added He, Ne, and Ar resembled that from a similar degree of KrF(B,C) quenching by added Kr.

C. Two-body quenching constants for Kr_2F^*

In order to have a strong Kr_2F^* fluorescence signal, all the Kr_2F^* quenching measurements were done with 300 Torr of Kr and 2 Torr of F_2. Since Kr_2F^* decays slowly, the waveform was recorded for 200 ns instead of the 100 ns interval used for the KrF(B,C) measurements. Although scattered laser light [and KrF(C–A) emission] always appeared with the Kr_2F^* waveform, the decay constant of Kr_2F^* could be obtained from analysis of the waveform after $\sim$30 ns [see Fig. 2(b)]. All the KrF(B,C) has disappeared after 30 ns and the formation rate for Kr_2F^* cannot interfere with the decay kinetics.

The quenching constant for Kr_2F^* by F_2 and the radiative lifetime have been measured by several workers. The early measurements by Quigley and Hughes[6] gave $k_{F_2} = (4.3 \pm 0.4) \times 10^{-10}$ cm^3 s^{-1} and $\tau_{Kr_2F^*} = 181 \pm 12$ ns. Morgan and Szoke[4] reported $k_{F_2} = 1.5 \times 10^{-10}$ cm^3 s^{-1}. The rate constant and lifetime determined by Ueda *et al.*[25] were 1.8×10^{-10} cm^3 s^{-1} and 210 ns. A previous report from our laboratory[16(b)] gave $\tau_{Kr_2F^*} = 240 \pm 35$ ns and $k_{F_2} = (1.5 \pm 0.2) \times 10^{-10}$ cm^3 s^{-1}. In order to confirm that our current analysis for the waveforms of Kr_2F^* was satisfactory, we repeated the k_{F_2} measurement. Figure 9 shows a plot of the first-order decay constant vs F_2 pressure; k_{F_2} obtained from the slope is $(1.2 \pm 0.2) \times 10^{-10}$ cm^3 s^{-1} and the intercept gives $\tau_{Kr_2F^*} = 220 \pm 20$ ns. Our new result supports the smaller range of values for k_{F_2}; the lifetime for Kr_2F^* seems established as 220 ns to within 10% uncertainty. In a solid Kr matrix, the Kr_2F^* radiative lifetime is 140 ns.[21]

The Stern–Volmer plots for He, Ne, and Ar with 300 Torr of Kr and 2 Torr of F_2 are shown in Fig. 10, and data for 300–600 Torr of Kr also are shown. Within the experimental uncertainty of the decay constants, there was no enhancement of the decay rate of Kr_2F^* by 400 Torr of added He, Ne, or Ar. Although there seems to be a slightly positive slope for some of the plots, direct comparison of the waveforms with and without 400 Torr of added He, Ne, and Ar showed that there was no detectable difference. Thus, the limit to the rate constants was set as 10% of the sum of the radiative decay and F_2 quenching rates, which gives $< 1.0 \times 10^{-13}$ cm^3 s^{-1} for the quenching constants of He, Ne, and Ar and $< 0.6 \times 10^{-13}$ cm^3 s^{-1} for Kr. Our data are consistent with other lower limits to these rate constants, which generally are in the range of 0.2×10^{-13} cm^3 s^{-1}.[6] The intercept values in Fig. 10 are consistent with the sum of $\tau_{Kr_2F^*}^{-1} + k_{F_2}(F_2)$ mentioned above.

The quenching rate constant of Kr_2F^* by Xe was measured again as part of our systematic study of Kr_2F^* in

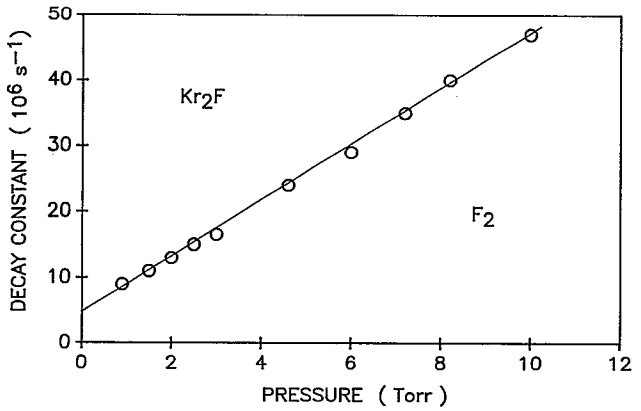


FIG. 9. Concentration dependence of the decay constants for Kr_2F^* for F_2 in 300 Torr of Kr. The intercept is $(4.5 \pm 0.5) \times 10^6$ s^{-1}, which corresponds to (220 ± 20) ns for the Kr_2F^* lifetime. The Kr_2F^* data shown in Figs. 9–12 were obtained by viewing the Kr_2F^* fluorescence with filters centered at 410 nm; 1000 laser shots usually were used to obtain a waveform.

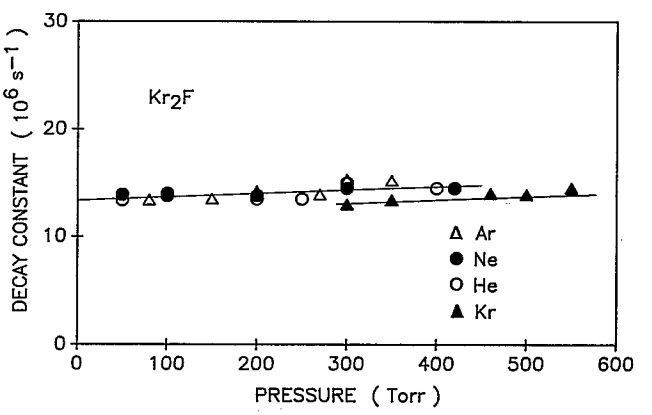


FIG. 10. Concentration dependence of the first-order decay constants of Kr_2F^* in He, Ne, and Ar with 300 Torr of Kr and 2 Torr of F_2. The decay constants for Kr_2F^* in Kr (up to 600 Torr) with 2 Torr of F_2 is also shown; see the text for how limits to the rate constants were assigned.

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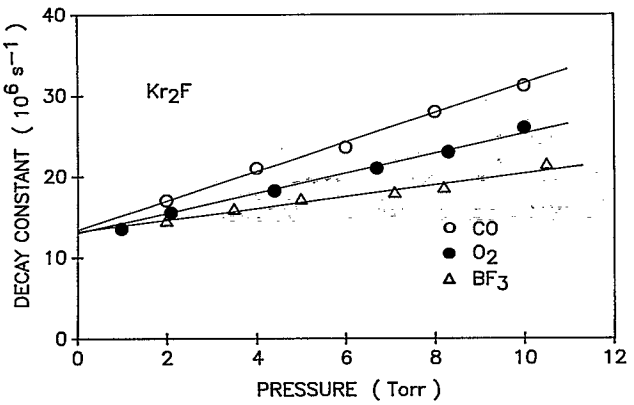


FIG. 11. Concentration dependence of the first-order decay constants of Kr_2F^* in CO, O_2, and BF_3 with 300 Torr of Kr and 2 Torr of F_2. These plots are representative for reagents with large rate constants.

rare gases. The new value for k_{Xe} is $(1.1\pm0.3)\times10^{-10}$ $cm^3\ s^{-1}$, which is somewhat smaller than the value previously reported from our laboratory $(1.5\pm0.3)\times10^{-10}$ $cm^3\ s^{-1}$,[16(b)] but the two results are within their combined error limits. A k_{Xe} value in the $1.0–1.5\times10^{-10}$ range is in accord with the measurements of Sauerbrey *et al.*[26] The large rate constant for Xe is consistent with a displacement mechanism for quenching of Kr_2F^*.

The quenching rate constants for Kr_2F^* were measured for the same molecular reagents as employed for KrF(*B,C*). Stern–Volmer plots that are representative of reagents with large and small quenching constants are shown in Figs. 11 and 12, and the values of the rate constants are summarized in Table I. The first-order decay constants declined by a factor of 3 for the reagents with larger constants and the plots should define reliable quenching constants. For N_2 and CF_4, the degree of quenching actually was insignificant, and we only can quote upper limits to those rate constants. A slight degree of quenching was found with SF_6 (see Fig. 12) and the rate constant is $(4.8\pm1.0)\times10^{-13}$ $cm^3\ s^{-1}$. Quenching also

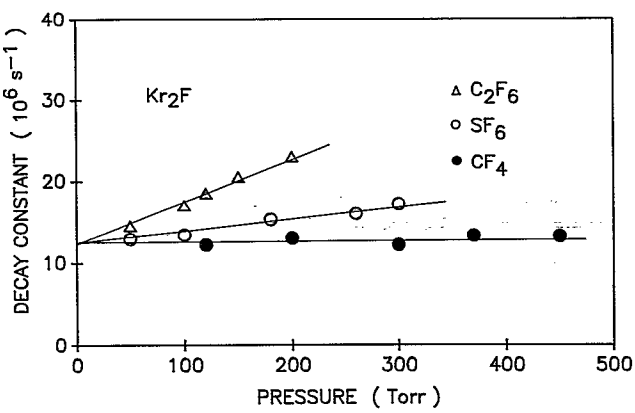


FIG. 12. Concentration dependence of the first-order decay constants of Kr_2F^* in C_2F_6, SF_6, and CF_4 in 300 Torr of Kr and 2 Torr of F_2. The quenching rate by CF_4 was too slow to measure.

was observed for C_2F_6; the waveforms were shown in Fig. 2(b) and the Stern–Volmer plot is in Fig. 12. As for KrF(*B,C*), the rate constant for C_2F_4 is larger than for C_2F_6. The quenching constants for Kr_2F^* are one order of magnitude smaller than those for KrF(*B,C*) for most reagents. However, the difference is two orders of magnitude or more for N_2, CF_4, and SF_6.

To our knowledge, quenching constants for Kr_2F^* have not previously been reported except for F_2, SF_6, and NF_3. The F_2 results were discussed above. The NF_3 quenching has been investigated in two independent experiments using electron beam pumped excitation.[19(b),27] We only included the smaller number[19(b)] in Table I because the values for SF_6 $(2.7\times10^{-9}$ $cm^3\ s^{-1})$, NF_3 $(6.0\times10^{-9}$ $cm^3\ s^{-1})$, and Kr$(6.0\times10^{-13}$ $cm^3\ s^{-1})$ from Ref. 27 are too large to be credible. Even the 5.1×10^{-11} $cm^3\ s^{-1}$ value[19(b)] given in Table I for NF_3 probably is too large and the radiative lifetime reported for Kr_2F^* in that work was too small by a factor of 2. Other species generated by the intense *e*-beam possibly caused deactivation of Kr_2F^* in addition to the quenching by NF_3.

IV. DISCUSSION

A. Comparison of rate constants with XeCl(*B,C*) and Xe_2Cl^*

Table I provides a set of quenching rate constants for KrF(*B,C*) and Kr_2F^* for a 300 K Boltzmann equilibrium mixture of KrF(*B*) and KrF(*C*) [K_{eq} is between 3 and 4 in favor of KrF(*C*)], and no distinction can be made between the individual *B* and *C* states with regard to quenching rates. Comparison with some of the high pressure steady-state sensitization results is worthwhile.[13] Based on the integrated *B–X* and *C–A* emission intensities in high pressures of CF_4, K_{eq} was estimated as 2–3 in approximate agreement with the value assigned from τ_{eff}.[16] In the sensitization experiments, the Kr pressure was very low (<0.1 Torr), so the mixed three-body rates involving Kr with other rare gases were negligible. Because of interference from the ArKrF* emission arising from the three-body ArKrF* formation in Ar, the best measurement for the I_B/I_C ratio was obtained with CF_4 as the buffer gas. Data could be obtained from up to 2000 Torr of CF_4, although the total KrF* fluorescence signal was reduced. The quenching constant in Table II for CF_4 predicts that the signal would be reduced by a factor of 40, which is in approximate agreement with the steady-state data. The three-body 2Ar+KrF(*B,C*) rate seemed to be larger than the 2Ar+KrF(*D*) rate, according to data from the steady-state experiments. Independent measurement of the KrF(*D*) quenching rates would be useful.

The KrF(*B*)↔KrF(*C*) collisional mixing rates are not fully characterized and the mixing, quenching, and vibrational relaxation rates may be competitive for some experimental conditions. Using photoassociation of Kr and F atoms to prepare KrF(*B*) in low *v* levels, Hakuta *et al.*[17] measured a $B\rightarrow C$ transfer constant of $1.2\pm0.3\times10^{-10}$ $cm^3\ s^{-1}$ for Kr. The rate constants[13] for the mixing of high vibrational levels of KrF(*B*) and KrF(*C*) by Ar, Ne, and

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He are $\sim 1\times10^{-10}$ $cm^3\,s^{-1}$, and Kvaran[12] has demonstrated that the *effective* vibrational relaxation rate constants are $3\text{–}40\times10^{-11}$ $cm^3\,s^{-1}$ for $v=13\text{–}40$ in He, Ne, and Ar. The vibrational relaxation rate constants for the lower vibrational levels become smaller in He, Ne, and Ar, just as for XeCl(B,C).[13] Care should be exercised in using the quenching constants of Table I for the quenching of either KrF(B) or KrF(C) alone.

Nearly all common tank gas impurities quench KrF(B,C) with large rate constants and this can be a practical concern for laser applications. Comparison with other experimental measurements of k_Q was made in preceding sections, and the quenching constants in Kr were discussed in the earlier report.[16(a)] A recent study[17] gives $k_{Kr}=7\pm3\times10^{-12}$ $cm^3\,s^{-1}$, which is larger than our value, but it was obtained by fixing the Kr three-body rate to a rather small value in the data analysis. Thus, the data would give a smaller k_{Kr} if a larger three-body constant was adopted. The only KrF(B,C) quenching constants for which we have some concern are those for He and NF_3; both possibly may be too large. The rate constants for He, Ne, Ar, Kr, N_2, and CF_4 with Kr_2F^* were too small to measure, and we quote only upper limits. The other three perfluorinated molecules had measurable quenching constants for Kr_2F^*; the difference between CF_4 and SiF_4 in terms of quenching rates seems especially noteworthy. The quenching constant for HF with Kr_2F^* seems small, but we have no reason to doubt the data.

The set of rate constants for KrF(B,C) and Kr_2F^* can be compared to those for XeCl(B,C) and Xe_2Cl^* and the ratios of the two-body rate constants are summarized in Table III. The former two molecules are 0.93 and $\sim$0.6 eV above the XeCl(B,C) and Xe_2Cl^* states, respectively. All four molecules are ionic in nature. A common feature of the three-body rates for KrF(B,C) and XeCl(B,C) is that the mixed three-body rates are more important than the homogeneous three-body rates, even for cases that can give bound trimers such as ArKrF* or KrXeF*, e.g., 2Ar +KrF(B,C) has a smaller rate constant than Ar+Kr +KrF(B,C). The quenching constants for Cl_2 and HCl with XeCl(B,C) are two times larger than the F_2 and HF constants for KrF(B,C). The two-body constants for XeCl(B,C) were divided[15] into two groups based on whether the rate constants were larger or smaller than 1×10^{-10} $cm^3\,s^{-1}$ (or the cross sections were larger or smaller than 20 $Å^2$). The small cross section group, which extends from He to NF_3 in Table III, showed a strong correlation between the magnitude of k_Q and the polarizability of the reagent. The k_{KrF^*}/k_{XeCl^*} ratio tends to be $\gtrsim2$ for this small cross section group and the ratio increases to 4–10 for the N_2–NF_3 subset. Of course, k_{Xe} is anomalous for KrF(B,C) because the displacement mechanism is much more facile than two-body physical quenching for XeCl(B,C). In this comparison, the large k_{NF_3} for KrF(B,C) actually seems normal. The group of molecules with large cross sections extends from CO through CF_3Cl in the list of Table III [the list could include several more molecules from Table I; but these were not studied with XeCl(B,C)]. The k_{KrF^*}/k_{XeCl^*} ratios are slightly less than unity for the group of molecules with large cross sections.

TABLE III. A comparison of two-body quenching rate constants for KrF(B,C) and $Kr_2F(4\,^2\Gamma)$ with XeCl(B,C) and $Xe_2Cl(4\,^2\Gamma)$.

Reagent	k_{KrF^*}/k_{XeCl^*}	$k_{Kr_2F^*}/k_{Xe_2Cl^*}$
He[a]	2.1	
Ne[a]	2.7	
Ar[a]	1.7	
Kr[a]	1.0	
Xe[a]	20	
N_2[b]	3.9	$\lesssim2.5$
CF_4[b]	10	~0.3[d]
SF_6[b]	4.8	~10
NF_3[b]	5.5	23
CO[b]	0.4	6.4
O_2[b]	0.8	8.0
CO_2[b]	0.5	0.9
N_2O[b]	0.2	3.3
CF_3Cl[c]	0.5	2.4
HF/HCl[e]	0.4	0.06
F_2/Cl_2[e]	0.5	0.3

[a]These XeCl(B,C) rate constants were taken from Ref. 14. The quenching constants for Xe_2Cl^* in Kr and Xe were less than 0.8×10^{-13} $cm^3\,s^{-1}$.
[b]These XeCl(B,C) and Xe_2Cl^* rate constants were taken from Ref. 15.
[c]Based on quenching rate constants for XeF* and XeCl*, CF_3H also could be added to the group with large cross sections.
[d]The quenching constant of Xe_2Cl^* by CF_4 probably is too large (Ref. 15).
[e]This comparison is for HF and F_2 with KrF* and Kr_2F^* and for HCl and Cl_2 with XeCl* and Xe_2Cl^*.

Neither Xe_2Cl^* nor Kr_2F^* seem to be quenched by rare gases (except for the displacement reaction between Xe and Kr_2F^*). In contrast to Ar_2F^*,[28] there is no evidence for the formation of Kr_3F^*, and Kr does not quench Kr_2F^* even at 600 Torr pressure. Also, there is no evidence for quenching of Kr_2F^* by Kr or Xe_2Cl^* by Xe in the condensed phase.[21,29,30] Except for CO_2 and the somewhat artificial comparison between F_2/Cl_2 and HF/HCl, the $k_{Kr_2F^*}/k_{Xe_2Cl^*}$ ratio is larger than unity for all molecules. Several molecules that were not studied with Xe_2Cl^* have quenching constants in the $2\text{–}4\times10^{-11}$ $cm^3\,s^{-1}$ range and Kr_2F^* is more readily quenched than Xe_2Cl^*.

B. Mechanisms for quenching KrF(B,C) and Kr_2F^*

A model to describe the quenching of KrF(B,C) or Kr_2F^* does not exist. Based on the generality of the rather facile two-body quenching and the calculated potentials for Xe+XeCl(B,C) and Kr+Kr(B,C),[30,31] we previously suggested that the mechanism for XeCl(B,C) involves a curve crossing between the $4\,^2\Gamma_Q$ potential correlating to XeCl(C)+Q and the $3\,^2\Gamma_Q$ potential correlating to XeCl(A)+Q. Since the two-body quenching constants of KrF(B,C) resemble those for XeCl(B,C), the basic mechanism must be the same as for XeCl(B,C). The long range part of the $4\,^2\Gamma_Q$ potentials for KrF*–Q and XeCl*–Q should be similar, since the main interactions are from ion–induced-dipole or ion–dipole forces; the repulsive limb will be at somewhat shorter distance for KrF*–Q than for XeCl*–Q. The covalent potential for KrF(A)–Q also will lie at a somewhat shorter range. These differences, plus the

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higher energy of KrF(B,C), seem to roughly cancel in terms of the expected crossing of the $4\,^2\Gamma_Q$ and $3\,^2\Gamma_Q$ potentials [see Fig. 8 of Refs. 15 and 31(a)]. The correlation of k_Q with polarizability of Q for the small cross section group probably is a consequence of the dependence of the attractive nature of the $4\,^2\Gamma_Q$ potential on the polarizability of the reagent. The $k_{XeF^*} < k_{XeCl^*} < k_{KrF^*}$ trend for those reagents in the small cross section group may be a consequence of more favorable crossing positions of the $3\,^2\Gamma$ and $4\,^2\Gamma$ potentials since the crossing position probably moves to smaller RgX*–Q distance as the upper state energy increases. Realistic calculations of model potentials are needed to test these ideas.

The magnitude of the cross section for molecules in the large rate constant group probably are largely determined by the orbiting criteria of the collision. The somewhat smaller dipole moment of KrF*, relative to XeCl*, may explain the slightly smaller k_{KrF^*} values for this group of reagents. The actual quenching mechanism would still be the curve crossing at short range mentioned above, but with an efficiency approaching unity.

The much smaller quenching cross sections for Xe_2Cl^* (and for Kr_2F^*) relative to XeCl(B,C) [and KrF(B,C)] imply a different quenching mechanism. We previously suggested that the mechanism was collisional coupling to the background of dissociative levels correlating to 2Xe+Cl (and to 2Kr+F).

V. CONCLUSIONS

The two-body quenching rate constants have been measured for a 300 K equilibrium mixture of KrF(B) and KrF(C) with 15 molecular reagents and for the rare gases. The values range from $2\text{–}4\times10^{-12}$ cm^3 s^{-1} for the rare gases and up to 2×10^{10} cm^3 s^{-1} for several molecular reagents and for Xe. The quenching constant for Xe is large because of the displacement mechanism giving XeF*. Some of the perfluorinated molecules have intermediate values for the quenching constants. The mixed three-body rate constants for He and Ar (with Kr) are in the 4–6 $\times10^{-31}$ cm^3 s^{-1} range and Kr_2F^* is the product. The two-body quenching constants for Kr_2F^* were measured for the same set of reagents. Quenching constants for the rare gases (except Xe), N_2 and CF_4 are $<1\times10^{-13}$ cm^3 s^{-1}. The quenching constants of Kr_2F^* by most molecules are one to two orders of magnitude smaller than for KrF(B,C). As for KrF(B,C), Xe quenches Kr_2F^* by a displacement mechanism giving XeF(B,C). The trends in the quenching rate constants for KrF(B,C) and $Kr_2F(4\,^2\Gamma)$ are very similar to those for XeCl(B,C) and $Xe_2Cl(4\,^2\Gamma)$, and the quenching mechanism for the Kr fluorides must be similar to that of the Xe chlorides. The radiative lifetime of Kr_2F^* was measured as 220 ± 20 ns, which is in modest agreement with the calculated value of 133 ns.[31(a)]

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